

Profit Borrowing Restraint and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Profit Borrowing Restraint (PBR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on PBR achieves an annualized gross (net) Sharpe ratio of 0.43 (0.31), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 18 (13) bps/month with a t-statistic of 2.53 (1.86), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, change in net operating assets, Asset growth) is 14 bps/month with a t-statistic of 2.10.

1 Introduction

Asset prices reflect investors' marginal utility of consumption across states and time, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While the consumption-based asset pricing paradigm provides a fundamental framework for understanding risk premia, empirical tests often struggle to identify firm characteristics that reliably capture variation in consumption risk exposure. We identify Profit Borrowing Restraint (PBR) as a novel predictor of stock returns that captures firms' sensitivity to aggregate consumption dynamics. The signal measures the extent to which firms rely on internal profit generation rather than external borrowing, reflecting their exposure to states where marginal utility is particularly high and external financing becomes constrained.

Profit Borrowing Restraint captures firms' exposure to consumption risk through their differential sensitivity to intertemporal marginal rates of substitution. Firms with high PBR generate profits internally without relying on external debt markets, making them more resilient during periods of high marginal utility when credit conditions tighten and consumption growth slows [Campbell and Cochrane \(1999\)](#). These firms provide insurance against consumption disasters and adverse macroeconomic shocks, commanding lower risk premia as investors value their state-contingent payoffs [Rietz \(1988\)](#); [Barro \(2006\)](#). The consumption-based interpretation builds on the long-run risks framework, where firms' differential exposure to persistent consumption shocks drives cross-sectional return variation [Bansal and Yaron \(2004\)](#).

The state-dependence of PBR's predictive power reflects its connection to time-varying risk premia through habit formation models. During bad times characterized by low surplus consumption ratios, investors become particularly risk-averse and demand higher compensation for bearing systematic risk [Campbell and Cochrane \(1999\)](#). Firms with low PBR—those heavily reliant on external borrowing—face amplified consumption risk exposure during these periods as their financing constraints

bind precisely when marginal utility is high [Gomes and Schmid \(2021\)](#). This mechanism generates countercyclical risk premia that vary with aggregate consumption dynamics and disaster probabilities [Wachter \(2013\)](#).

The cross-sectional variation in PBR directly maps to heterogeneous loadings on consumption-based factors through firms’ production technologies and financing decisions. High PBR firms maintain operational flexibility that allows them to smooth dividends across consumption states, reducing their covariance with the stochastic discount factor [Zhang \(2005\)](#); [Liu et al. \(2009\)](#). The resulting dispersion in consumption betas generates the return predictability we document, as markets rationally price firms’ differential exposure to aggregate consumption risk through the intertemporal marginal rate of substitution [Lettau and Ludvigson \(2001b\)](#).

We find strong evidence that Profit Borrowing Restraint predicts cross-sectional stock returns, with a value-weighted long-short strategy achieving an annualized gross Sharpe ratio of 0.43. The strategy generates a monthly average excess return of 21 basis points with a t-statistic of 3.09, demonstrating both statistical and economic significance. The alpha relative to the Fama-French six-factor model equals 18 basis points per month with a t-statistic of 2.53, indicating that PBR captures a distinct dimension of systematic risk not spanned by existing factors.

The predictive power of PBR remains robust across various portfolio construction methodologies and size groups. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the PBR strategy delivers an average return of 27 basis points per month with a t-statistic of 3.16. The strategy’s factor loadings reveal a significant positive exposure to the CMA factor of 0.25 with a t-statistic of 5.21, consistent with PBR capturing firms’ investment restraint during periods of high marginal utility. After accounting for transaction costs using high-frequency effective spreads, the strategy maintains a net Sharpe ratio of 0.31, ranking in the top 8% of anomalies in the factor zoo.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the PBR strategy retains an alpha of 14 basis points per month with a t-statistic of 2.10. The signal’s incremental predictive power persists in Fama-MacBeth regressions controlling for related financing and investment characteristics, though the coefficient becomes statistically insignificant when jointly controlling for all six related signals. The PBR strategy improves the mean-variance efficient frontier for investors with access to standard factor models, with positive net generalized alphas across all five factor model specifications tested.

Our paper contributes to the consumption-based asset pricing literature by identifying a firm characteristic that captures cross-sectional variation in consumption risk exposure. While [Lettau and Ludvigson \(2001a\)](#) and [Parker and Julliard \(2005\)](#) establish the importance of consumption risk for aggregate market returns, we demonstrate that PBR proxies for firm-level differences in consumption beta that generate predictable return variation. Our findings complement [Yogo \(2006\)](#) who shows that durable consumption risk explains the cross-section, by focusing on how firms’ financing constraints interact with consumption dynamics to determine risk premia.

We extend the investment-based asset pricing literature by linking profit generation and borrowing decisions to consumption risk through the stochastic discount factor. Unlike [Cooper et al. \(2008\)](#) who document that asset growth predicts returns through overinvestment, and [Lyandres et al. \(2008\)](#) who examine external financing effects, we show that the interaction between profitability and borrowing restraint captures exposure to marginal utility states. Our PBR measure provides incremental explanatory power beyond these related anomalies, with a significant alpha of 14 basis points when controlling for asset growth, net external financing, and change in net operating assets simultaneously.

Methodologically, we employ the rigorous testing framework of [Novy-Marx and Velikov \(2023b\)](#) to establish the robustness of PBR’s return predictability, addressing

concerns about data mining and publication bias in the anomalies literature raised by [Harvey et al. \(2016\)](#). The signal’s performance net of transaction costs, its effectiveness among large-cap stocks, and its ability to expand the mean-variance efficient frontier demonstrate genuine economic content beyond statistical artifact. These findings have important implications for understanding how firms’ operational and financing decisions determine their systematic risk exposures in a consumption-based framework, providing new evidence on the fundamental sources of cross-sectional return variation.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on non-financial firms and exclude utilities following standard practice in the asset pricing literature. Profit Borrowing Restraint signal requires two key variables from firms’ financial statements. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing firms’ external financing activities through debt markets. Gross profit (GP) measures the difference between total revenues and cost of goods sold, representing the fundamental profitability of a firm’s operations before operating expenses, interest, and taxes. Both variables are reported in millions of dollars and reflect annual flow measures from firms’ cash flow statements and income statements, respectively. We construct the Profit Borrowing Restraint signal by computing the year-over-year change in long-term debt issuance scaled by lagged gross profit: $(DLTIS(t) - DLTIS(t-1)) / GP(t-1)$. This ratio captures the change in a firm’s debt financing activity relative to its profit-generating capacity. A higher

value indicates that a firm is accelerating its debt issuance relative to its profitability, potentially signaling either growth opportunities requiring external financing or deteriorating internal cash generation necessitating increased borrowing. We calculate this measure annually using fiscal year-end values and require non-missing values for both DLTIS and GP in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with non-positive gross profit in the denominator are excluded from the analysis to ensure meaningful economic interpretation of the ratio.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the PBR signal. Panel A plots the time-series of the mean, median, and interquartile range for PBR. On average, the cross-sectional mean (median) PBR is -0.14 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input PBR data. The signal's interquartile range spans -0.31 to 0.36. Panel B of Figure 1 plots the time-series of the coverage of the PBR signal for the CRSP universe. On average, the PBR signal is available for 6.14% of CRSP names, which on average make up 7.43% of total market capitalization.

4 Does PBR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on PBR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high PBR portfolio and sells the low PBR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model

(FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short PBR strategy earns an average return of 0.21% per month with a t-statistic of 3.09. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.18% to 0.26% per month and have t-statistics exceeding 2.53 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.25, with a t-statistic of 5.21 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 550 stocks and an average market capitalization of at least \$1,285 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for

the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.87. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -12-17bps/month. The lowest return, (-12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -2.02. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the PBR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in fourteen cases.

Table 3 provides direct tests for the role size plays in the PBR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and PBR, as well as average returns and alphas for long/short trading PBR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the PBR strategy achieves an average return of 27 bps/month with a t-statistic of 3.16. Among these large cap stocks, the alphas for the PBR strategy relative to the five most common factor models range from 23 to 31 bps/month with t-statistics between 2.57 and 3.57.

5 How does PBR perform relative to the zoo?

Figure 2 puts the performance of PBR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the PBR strategy falls in the distribution. The PBR strategy’s gross (net) Sharpe ratio of 0.43 (0.31) is greater than 83% (92%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the PBR strategy (red line).² Ignoring trading costs, a \$1 invested in the PBR strategy would have yielded \$2.29 which ranks the PBR strategy in the top 9% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the PBR strategy would have yielded \$1.31 which ranks the PBR strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the PBR relative to those. Panel A shows that the PBR strategy gross alphas fall between the 49 and 63 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43%

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The PBR strategy has a positive net generalized alpha for five out of the five factor models. In these cases PBR ranks between the 66 and 78 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does PBR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of PBR with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price PBR or at least to weaken the power PBR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of PBR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PBR}PBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on PBR. Stocks are finally grouped into five PBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PBR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on PBR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the PBR signal in these Fama-MacBeth regressions exceed -0.41, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on PBR is -0.39.

Similarly, Table 5 reports results from spanning tests that regress returns to the PBR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the PBR strategy earns alphas that range from 15-18bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.14, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the PBR trading strategy achieves an alpha of 14bps/month with a t-statistic of 2.10.

7 Does PBR add relative to the whole zoo?

Finally, we can ask how much adding PBR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the PBR signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes PBR grows to \$1208.74.

8 Conclusion

This study provides compelling evidence that Profit Borrowing Restraint (PBR) represents a economically meaningful and statistically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that PBR-based trading strategies generate substantial risk-adjusted returns even after accounting for transaction costs and controlling for well-established risk factors.

The key findings of our research underscore the robustness of PBR as a return predictor. The value-weighted long/short strategy delivers an annualized net Sharpe

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which PBR is available.

ratio of 0.31, which, while reduced from the gross Sharpe ratio of 0.43 due to implementation costs, remains economically significant. More importantly, the strategy generates positive and statistically significant alphas relative to the Fama-French five-factor model plus momentum, with net abnormal returns of 13 basis points per month (t-statistic = 1.86). The signal’s persistence is further validated by its ability to generate a significant alpha of 14 basis points per month (t-statistic = 2.10) even after controlling for six closely related financing and investment factors, including changes in financial liabilities, net debt financing, net external financing, inventory growth, changes in net operating assets, and asset growth.

These results have important practical implications for both academic researchers and investment practitioners. For academics, PBR adds to our understanding of how financing constraints and profit dynamics interact to influence expected returns, contributing to the broader literature on the intersection of corporate finance and asset pricing. For practitioners, the signal’s ability to generate positive risk-adjusted returns net of transaction costs suggests potential value in incorporating PBR-based screens into quantitative investment strategies, particularly as part of a multi-factor approach.

However, several limitations of this study warrant acknowledgment. First, while we follow the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and market conditions examined may not fully capture the signal’s performance across all market regimes. Second, the reduction in Sharpe ratio from 0.43 to 0.31 after accounting for transaction costs highlights the importance of implementation efficiency, and actual trading costs may vary substantially across different market participants and time periods. Third, although we control for several related factors, the possibility of omitted risk factors or alternative explanations for the PBR premium cannot be entirely ruled out.

Future research should explore several promising avenues. First, investigating

the economic mechanisms underlying the PBR effect could provide deeper insights into why this signal predicts returns. Are firms with high profit borrowing restraint systematically undervalued due to investor inattention, or does the signal capture a risk premium associated with financing constraints? Second, examining the international evidence on PBR would help establish whether this is a global phenomenon or specific to certain markets. Third, analyzing the signal’s performance during different macroeconomic conditions, particularly during credit crunches and expansionary periods, could reveal important time-variation in its effectiveness. Finally, exploring potential enhancements to the PBR signal through machine learning techniques or combination with other predictors may yield strategies with even stronger risk-adjusted returns.

In conclusion, this paper establishes Profit Borrowing Restraint as a robust and economically significant predictor of cross-sectional stock returns that survives stringent empirical tests and contributes meaningfully to our understanding of the determinants of expected returns in equity markets.

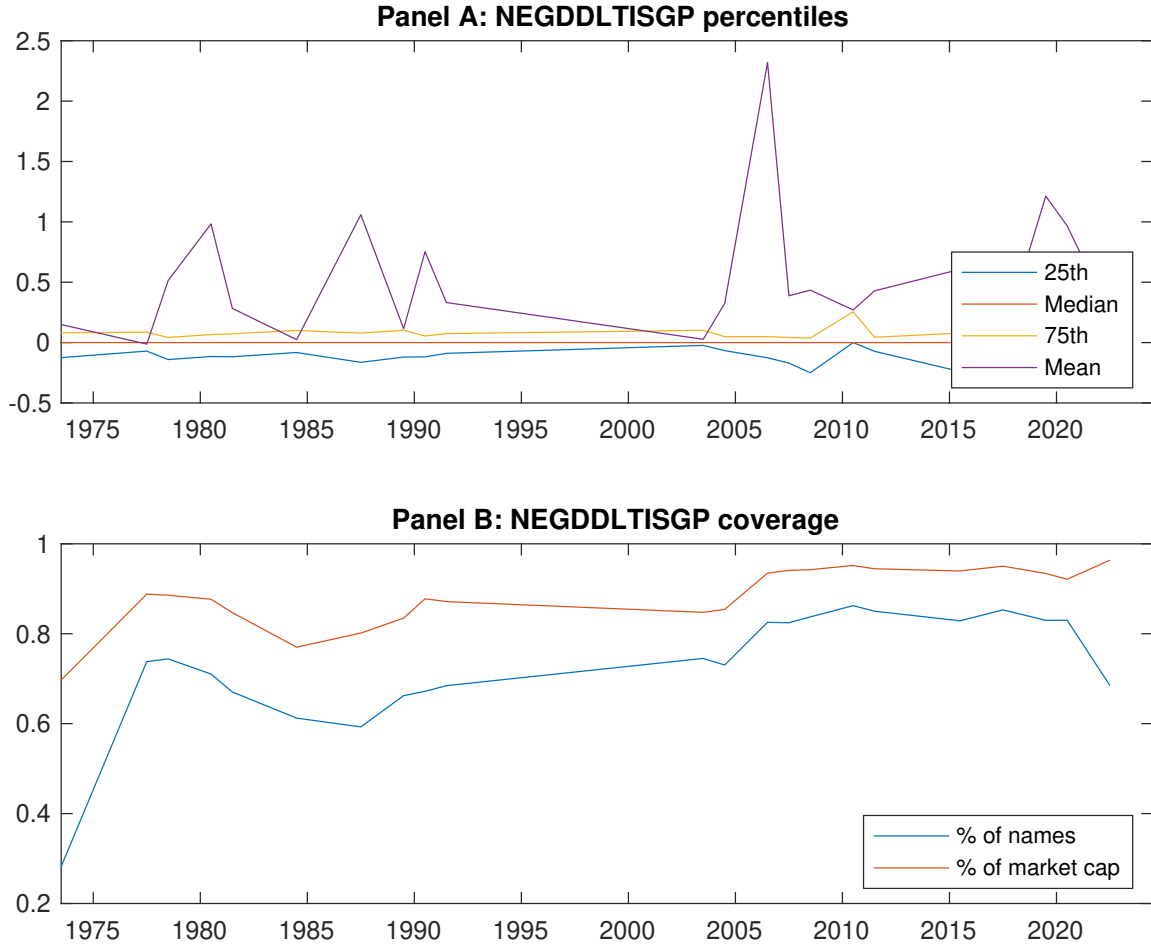


Figure 1: Times series of PBR percentiles and coverage.
This figure plots descriptive statistics for PBR. Panel A shows cross-sectional percentiles of PBR over the sample. Panel B plots the monthly coverage of PBR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on PBR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on PBR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.57 [2.65]	0.62 [3.50]	0.67 [3.43]	0.75 [4.25]	0.78 [3.86]	0.21 [3.09]
α_{CAPM}	-0.16 [-2.70]	0.02 [0.42]	0.01 [0.17]	0.15 [3.41]	0.10 [1.70]	0.25 [3.72]
α_{FF3}	-0.21 [-3.76]	-0.02 [-0.42]	0.08 [1.51]	0.15 [3.54]	0.05 [0.89]	0.26 [3.75]
α_{FF4}	-0.19 [-3.36]	0.02 [0.34]	0.11 [2.20]	0.12 [2.74]	0.04 [0.65]	0.23 [3.24]
α_{FF5}	-0.17 [-2.97]	-0.07 [-1.63]	0.09 [1.73]	0.08 [1.77]	0.03 [0.48]	0.19 [2.81]
α_{FF6}	-0.16 [-2.76]	-0.04 [-0.96]	0.12 [2.25]	0.05 [1.29]	0.02 [0.35]	0.18 [2.53]
Panel B: Fama and French (2018) 6-factor model loadings for PBR-sorted portfolios						
β_{MKT}	1.09 [82.85]	0.97 [94.40]	0.98 [81.73]	0.97 [98.59]	1.05 [79.89]	-0.04 [-2.77]
β_{SMB}	0.07 [3.68]	-0.11 [-7.24]	-0.01 [-0.66]	-0.04 [-2.88]	0.13 [6.47]	0.05 [2.24]
β_{HML}	0.18 [7.24]	0.11 [5.36]	-0.18 [-7.71]	-0.06 [-3.01]	0.06 [2.54]	-0.12 [-3.84]
β_{RMW}	-0.01 [-0.43]	0.13 [6.48]	0.01 [0.60]	0.09 [4.79]	-0.00 [-0.09]	0.01 [0.28]
β_{CMA}	-0.15 [-3.82]	0.05 [1.59]	-0.05 [-1.41]	0.16 [5.53]	0.10 [2.58]	0.25 [5.21]
β_{UMD}	-0.02 [-1.26]	-0.05 [-4.70]	-0.04 [-3.58]	0.03 [3.32]	0.01 [0.86]	0.03 [1.72]
Panel C: Average number of firms (n) and market capitalization (me)						
n	634	550	1046	600	614	
me (\$10 ⁶)	1324	3129	2360	3237	1285	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the PBR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21 [3.09]	0.25 [3.72]	0.26 [3.75]	0.23 [3.24]	0.19 [2.81]	0.18 [2.53]
Quintile	NYSE	EW	0.13 [2.87]	0.15 [3.23]	0.14 [3.11]	0.15 [3.22]	0.13 [2.85]	0.14 [3.04]
Quintile	Name	VW	0.18 [2.58]	0.22 [3.29]	0.23 [3.43]	0.20 [2.90]	0.16 [2.33]	0.14 [2.06]
Quintile	Cap	VW	0.20 [3.06]	0.23 [3.56]	0.24 [3.76]	0.21 [3.16]	0.17 [2.66]	0.15 [2.33]
Decile	NYSE	VW	0.24 [2.78]	0.28 [3.32]	0.26 [3.08]	0.25 [2.84]	0.20 [2.33]	0.20 [2.25]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.16 [2.24]	0.19 [2.80]	0.20 [2.83]	0.18 [2.59]	0.14 [2.06]	0.13 [1.86]
Quintile	NYSE	EW	-0.12 [-2.02]					
Quintile	Name	VW	0.12 [1.75]	0.17 [2.46]	0.17 [2.54]	0.15 [2.27]	0.11 [1.66]	0.10 [1.43]
Quintile	Cap	VW	0.15 [2.29]	0.18 [2.85]	0.19 [3.00]	0.17 [2.69]	0.14 [2.12]	0.12 [1.86]
Decile	NYSE	VW	0.17 [1.93]	0.20 [2.36]	0.19 [2.15]	0.18 [2.05]	0.13 [1.49]	0.12 [1.42]

Table 3: Conditional sort on size and PBR

This table presents results for conditional double sorts on size and PBR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on PBR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high PBR and short stocks with low PBR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	PBR Quintiles					PBR Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.67 [2.43]	0.91 [3.30]	1.03 [3.79]	1.02 [3.57]	0.72 [2.66]	0.06 [0.78]	0.06 [0.87]	0.06 [0.83]	0.06 [0.83]	0.04 [0.50]	0.04 [0.58]
	(2)	0.76 [2.89]	0.85 [3.37]	0.89 [3.54]	0.91 [3.64]	0.83 [3.29]	0.07 [0.95]	0.11 [1.45]	0.08 [1.09]	0.07 [0.94]	0.03 [0.38]	0.03 [0.37]
	(3)	0.79 [3.25]	0.79 [3.50]	0.81 [3.37]	0.89 [3.92]	0.85 [3.69]	0.06 [0.82]	0.10 [1.39]	0.11 [1.47]	0.10 [1.36]	0.10 [1.32]	0.10 [1.27]
	(4)	0.69 [3.10]	0.81 [3.82]	0.88 [3.95]	0.79 [3.75]	0.83 [3.83]	0.13 [1.70]	0.15 [1.98]	0.15 [1.87]	0.12 [1.53]	0.11 [1.40]	0.10 [1.17]
	(5)	0.50 [2.49]	0.59 [3.37]	0.59 [3.02]	0.67 [3.72]	0.77 [3.99]	0.27 [3.16]	0.30 [3.42]	0.31 [3.57]	0.29 [3.25]	0.24 [2.68]	0.23 [2.57]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	PBR Quintiles					PBR Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	383	385	384	385	381	36	33	31	33	34	
	(2)	106	106	106	106	106	60	60	59	60	60	
	(3)	76	76	76	76	76	108	108	104	107	106	
	(4)	64	64	64	64	64	228	240	230	235	227	
(5)	59	59	59	59	59	1320	2184	2013	2220	1439		

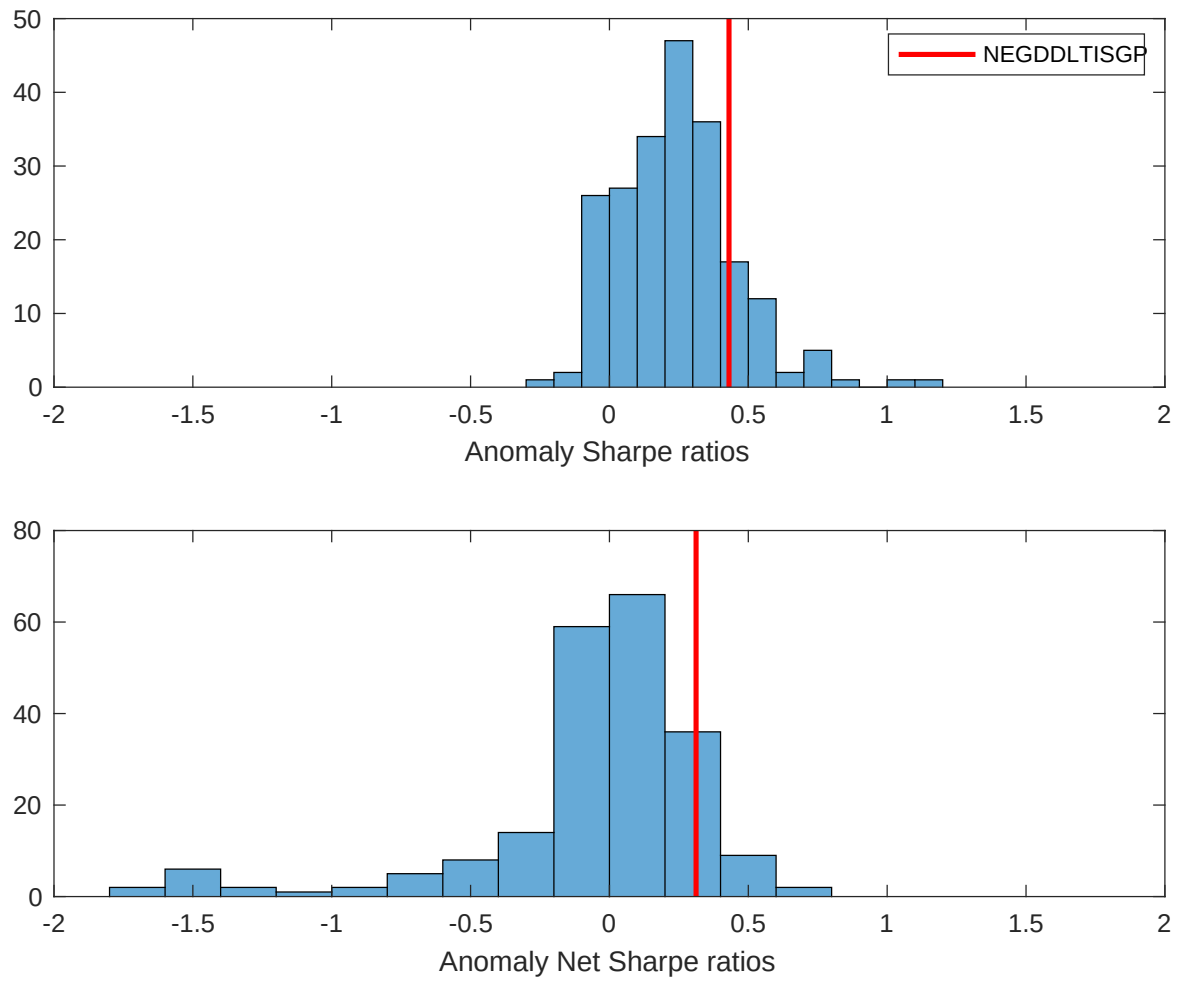


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the PBR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

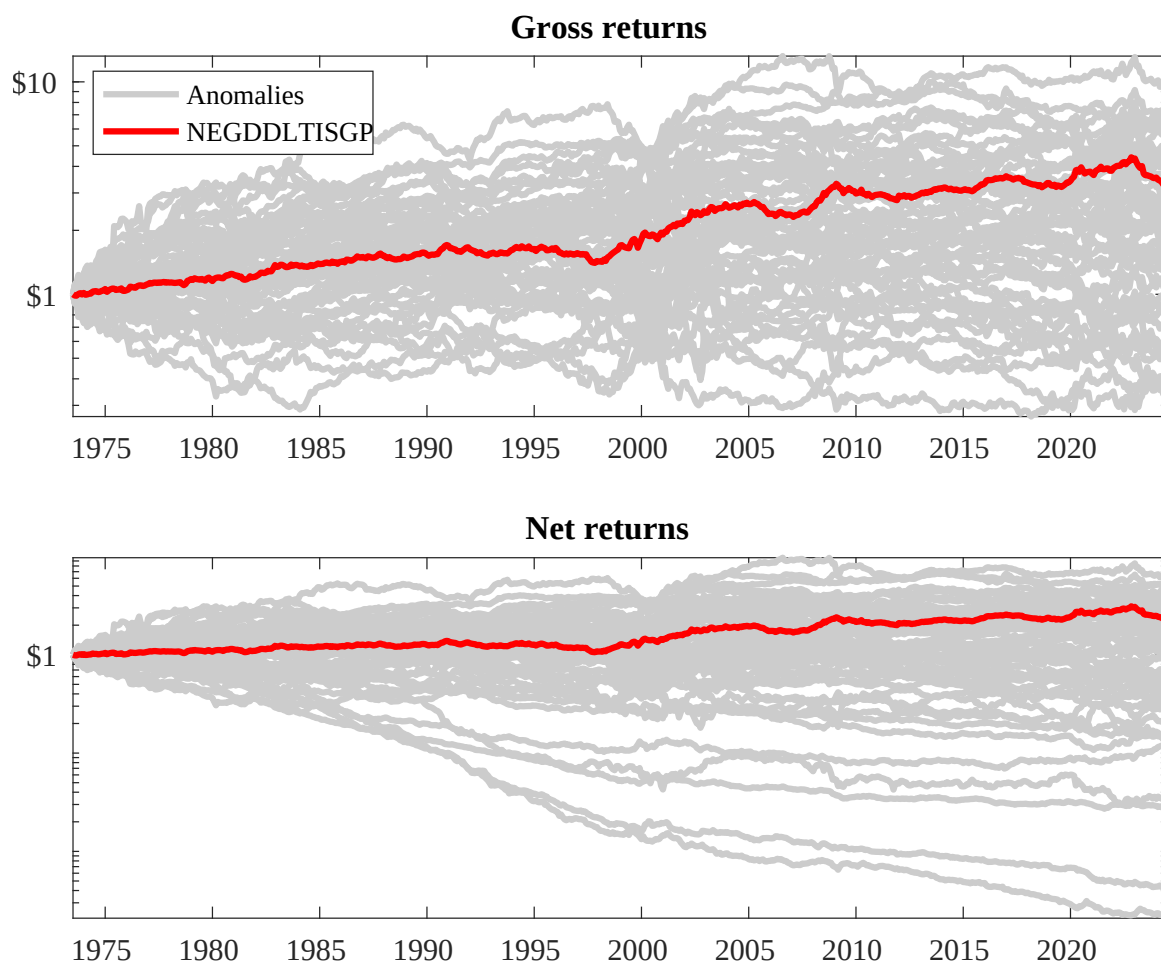
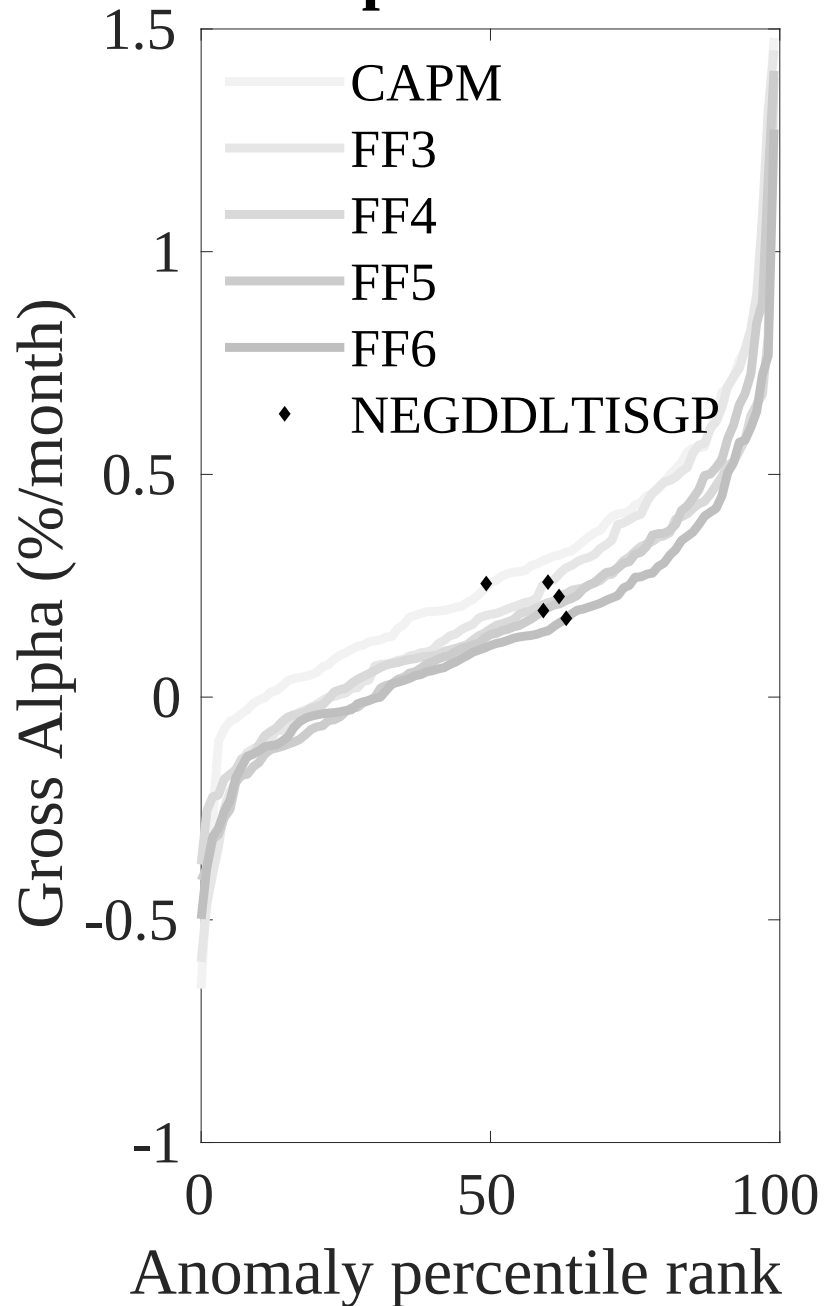


Figure 3: Dollar invested.

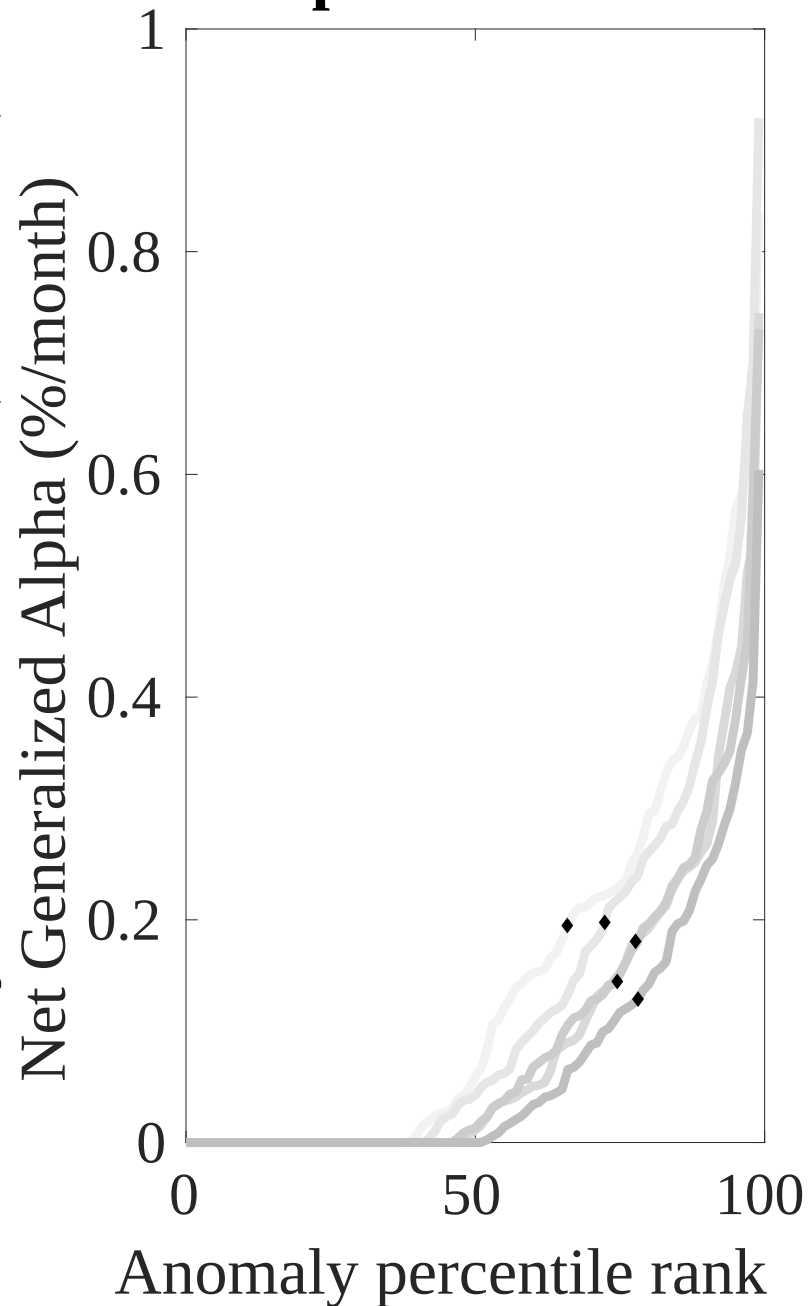
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the PBR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



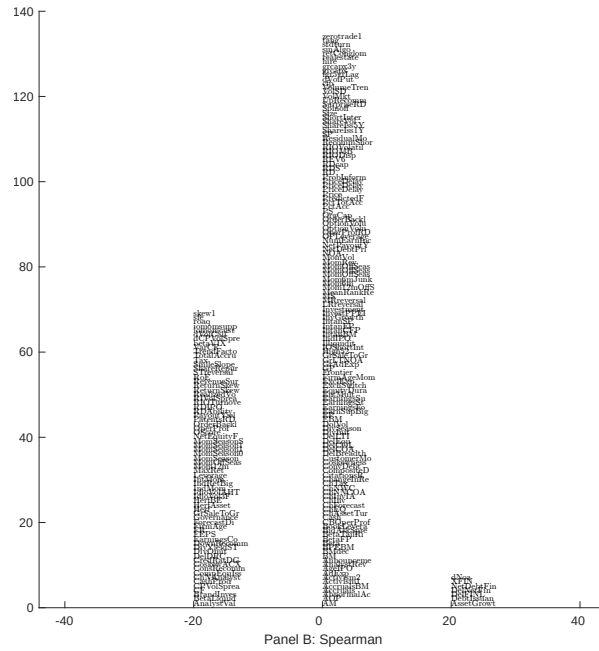
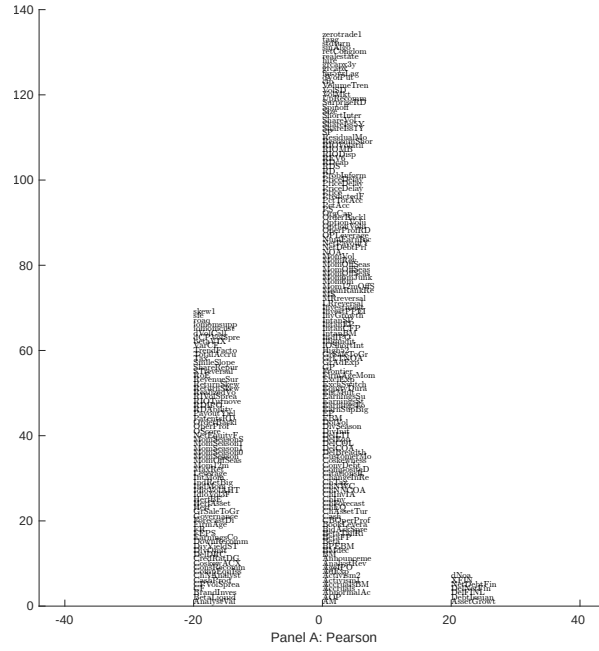


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with PBR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

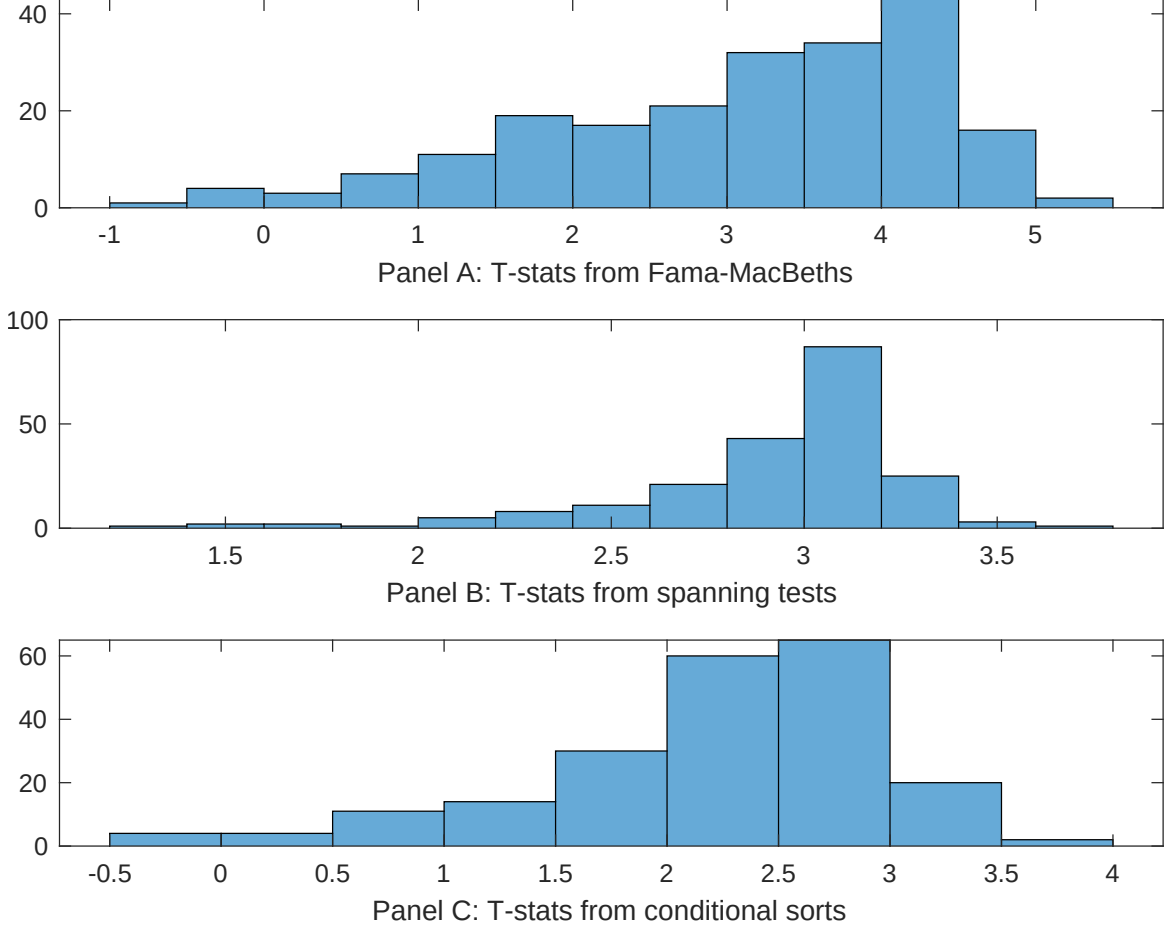


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of PBR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PBR}PBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on PBR. Stocks are finally grouped into five PBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PBR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on PBR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{PBR}PBR_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, change in net operating assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.44]	0.13 [5.40]	0.14 [5.70]	0.14 [5.32]	0.14 [5.75]	0.14 [5.88]	0.15 [5.71]
PBR	0.23 [0.13]	0.59 [0.33]	0.29 [1.58]	0.82 [3.49]	-0.69 [-0.41]	-0.14 [-0.08]	-0.97 [-0.39]
Anomaly 1	0.18 [9.63]						-0.11 [-2.23]
Anomaly 2		0.21 [9.73]					0.10 [1.52]
Anomaly 3			0.20 [6.56]				0.96 [1.64]
Anomaly 4				0.41 [7.03]			0.38 [0.69]
Anomaly 5					0.15 [10.16]		0.94 [5.21]
Anomaly 6						0.11 [8.77]	0.42 [1.84]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the PBR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{PBR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, change in net operating assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.15 [2.14]	0.16 [2.27]	0.16 [2.23]	0.18 [2.54]	0.16 [2.26]	0.17 [2.44]	0.14 [2.10]
Anomaly 1	24.32 [6.27]						18.89 [3.54]
Anomaly 2		22.46 [5.98]					7.44 [1.46]
Anomaly 3			14.40 [4.13]				11.20 [3.05]
Anomaly 4				7.24 [2.65]			6.55 [2.39]
Anomaly 5					8.54 [2.12]		-3.22 [-0.73]
Anomaly 6						5.21 [1.27]	-1.44 [-0.33]
mkt	-3.90 [-2.47]	-4.42 [-2.80]	-2.49 [-1.49]	-4.60 [-2.85]	-4.42 [-2.73]	-4.47 [-2.76]	-2.59 [-1.58]
smb	2.53 [1.05]	3.04 [1.26]	9.66 [3.62]	5.83 [2.37]	5.18 [2.11]	4.66 [1.88]	6.79 [2.45]
hml	-10.97 [-3.60]	-11.66 [-3.83]	-10.97 [-3.52]	-12.80 [-4.11]	-13.31 [-4.24]	-12.80 [-4.08]	-9.68 [-3.15]
rmw	-1.27 [-0.41]	-1.55 [-0.50]	-7.69 [-2.04]	1.97 [0.62]	1.06 [0.34]	0.94 [0.30]	-7.40 [-1.98]
cma	17.45 [3.65]	19.76 [4.18]	16.19 [3.11]	19.14 [3.60]	19.29 [3.46]	19.26 [2.81]	8.37 [1.22]
umd	0.57 [0.36]	1.23 [0.78]	2.58 [1.62]	1.94 [1.19]	2.39 [1.48]	2.79 [1.73]	-0.12 [-0.08]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	14	13	11	9	9	8	16

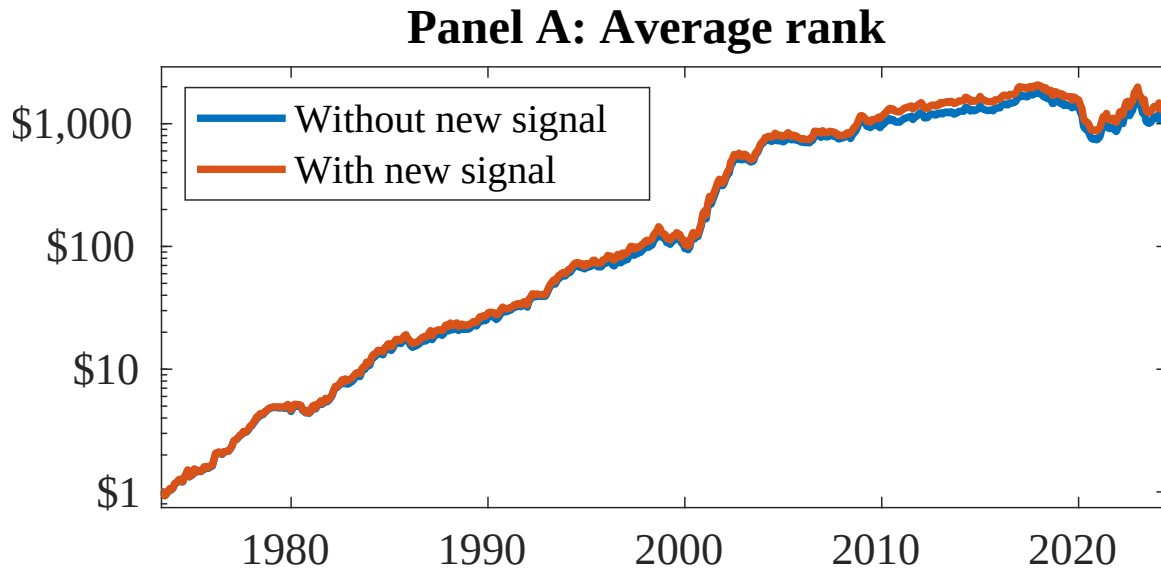


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as PBR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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